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Has the shortened drug distribution chain cut drug prices? Evidence from the Two-Invoice System in China☆

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ABSTRACT

Global concerns over rising drug prices have led to regulatory efforts in the pharmaceutical industry targeting price transparency and distribution efficiency. This study examines the impact of China's Two-Invoice System (TIS), a 2016 reform aimed at reducing drug costs by streamlining the pharmaceutical distribution chain and enhancing price transparency. With a theoretical model, we examine the decision-making for pharmaceutical manufacturers and distributors on pricing and promotion activities. We predict competing effects on drug prices: while intermediary markups inflated by double marginalization are reduced, the removal of efficient distributors may elevate supply chain expenses. Using a staggered difference-in-differences approach with procurement data from 2015 to 2019, we find that the implementation of TIS led to a 1.9% increase in average drug prices, contrary to policy expectations. Price increases were more pronounced for lower-priced drugs and in wealthier regions, for which the supply chain markups are limited and the effect of efficiency disruption might dominate. We also document a significant increase in sales and marketing expenses of the manufacturers, who are supposed to have absorbed a considerable amount of promotional costs previously borne by distributors. These findings underscore the unintended inefficiencies of regulatory reforms, highlighting the need for a balanced policy design that considers both cost control and market dynamics in healthcare.

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1. Introduction

The escalating healthcare costs have become a pressing global challenge, with pharmaceutical expenditures as a major driver (Baltagi et al., 2017; Ben-Aharon et al., 2017). In 2021, pharmaceutical costs accounted for 14.02% of total healthcare spending in the U.S. (Tichy et al., 2021) and 17% across OECD countries, while in China, drug expenses made up 37.5% of outpatient costs and 25.1% of inpatient costs.[2] Studies report an essential-medicine affordability index of 0.88 in China (Liu et al., 2024), near the WHO upper bound.[3] This growing burden all over the world has led to regulatory interventions aimed at curbing drug prices (Garber and Skinner, 2008; Chandra and Skinner, 2012; Kyle and Williams, 2017). Across countries, policy levers to restrict intermediaries' markups are worth considering, while such interventions may potentially risk disrupting supply chain efficiency.

In China, the Two-Invoice System (TIS), introduced in 2016, offers a salient policy experiment.[4] TIS was designed to streamline the pharmaceutical supply chain by restricting the number of intermediaries between drug manufacturers and hospitals, thereby improving transparency and containing inflated drug prices. The TIS mechanisms we study, removing stacked markups versus edging out the efficient distributors, travel beyond China and speak to ongoing global debates on drug price regulation and supply-chain reforms.

Previous studies have shown that a significant portion of drug prices in many countries is captured by intermediaries in the distribution chain (Kanavos et al., 2011; Aitken, 2016; Lee et al., 2020). In the U.S. and Europe, distributors retain around 20%–25% of the retail drug price (Sood et al., 2017; EFPIA, 2022). This phenomenon is particularly pronounced in China, the Chinese pharmaceutical distribution sector absorbs up to 53%–55%.[5] Given that pharmaceuticals constitute one-third of medical costs in China, this implies that nearly 18% of healthcare spending, or 1.3% of China's GDP, is attributed to the pharmaceutical distribution sectors. Thus, a policy tackling issues of this scale merits a comprehensive evaluation.

The TIS in China's pharmaceutical distribution sector provides a unique market setting to examine how a regulatory policy that rigidly changes market structure reshapes market outcome, which sheds light on the existing literature about the arrangement of a cost-saving distribution system (Liu et al., 2009; Oguoma et al., 2010; Wang et al., 2017; Joosse et al., 2023). It also offers a clean policy laboratory for a general question central to reforms elsewhere: While regulatory suppression of intermediaries' profits may reduce retail prices by alleviating double marginalization, it remains unclear whether such reductions sufficiently offset the upward price pressure induced by efficiency losses among intermediaries or by increased market power. These considerations extend the relevance of our analysis beyond China: from U.S. pharmacy benefit managers (PBMs) reforms[6] to European margin regulation[7] and low and middle-income countries (LMICs) public channels,[8] removing layers may simply reallocate market power unless pass-through and service-coverage safeguards are built into policy design.

Before the implementation of the TIS, China's pharmaceutical distribution was characterized by a complex, multi-tiered structure extending from manufacturers through various levels of wholesalers to hospitals. This network ensured broad availability across a vast and heterogeneous geography but also carried risks—counterfeit circulation, commercial bribery, opaque cost management, and multiple markups along the chain (Yu et al., 2010; Hasan et al., 2019; Ran et al., 2022). As depicted in Fig. 1, TIS streamlined the distribution process by imposing only one invoice from manufacturers to distributors and another one from distributors to hospitals, introducing a paradigm shift towards a single-tier distribution model.

A direct consequence of this policy is a noticeable shortening of the supply chains, as smaller entities that were struggling to compete were forced to exit supply chains to public hospitals. Administrative mandates by TIS create two opposing forces at play. First, the shift to a simplified single-tier distribution model aims to reduce transactional costs by eliminating multiple intermediary layers and improving transparency, potentially reducing multilevel markups and lowering drug prices (Berto Villas-Boas, 2007; Gil, 2015; Crawford et al., 2018; Rubens, 2023). Second, this streamlining suffers from the risk of price increases due to loss of efficiency from shortened supply chains, especially if smaller distributors with local information and networks are edged out. As illustrated in Fig. 2, after the TIS, larger distributors have increased their share of operating revenue, suggesting that consolidation of the distribution sector favors larger state-owned distributors. Given these competing forces, the overall effect of TIS on drug pricing

[2] See *China Health and Healthcare Development Statistics Bulletin*, National Health Commission, July 12, 2022, available at: https://www.nhc.gov.cn/wjw/c100381/202207/6aa5c4c7db614cceab727599675a4929.shtml (Accessed December 29, 2025).

[3] Drug affordability is defined as the total cost of a drug for treatment of a disease at a standard dose within a course of treatment divided by the minimum daily wage of a non-skilled person in a government department. It is equivalent to the number of working days required for a person to afford the drug costs.

[4] China has implemented several policies in addressing drug prices, including the Zero-Markup Drug Policy (ZMDP) in 2015, the TIS in 2016, and the Centralized Volume-Based Procurement (CVBP) in 2019. While the impacts of the ZMDP and CVBP have been well studied (Peng et al., 2021; Shi et al., 2019, 2023; Yi et al., 2015; Fang et al., 2021; Liu et al., 2024; Cao et al., 2024), the effects of the TIS remain less understood.

[5] Based on Shen and Liang (2020), 75% of retail prices in China are directed to the distribution channel. After accounting for approximately 15% in taxes and an estimated 5%–7% in delivery costs, approximately 53%–55% of the retail price is absorbed by the distribution sector.

[6] In the United States, policy debates and enforcement focus on PBMs and large distributors—asking whether intermediary scale lowers net acquisition costs yet raises patient or plan spending through limited pass-through, steering, or spread pricing. Consistent with these concerns, federal rulemaking has shifted pharmacy price concessions to the point of sale to improve patient-facing pass-through.

[7] In Europe, regulated distribution margins—often a sizable share of retail medicine prices—underscore that policy design, not only market structure, shapes incidence along the chain.

[8] In LMICs, public channels refer to government-managed procurement and distribution systems in which the state exercises direct oversight and control over intermediary activities to control costs while ensuring access, affordability, and service coverage.

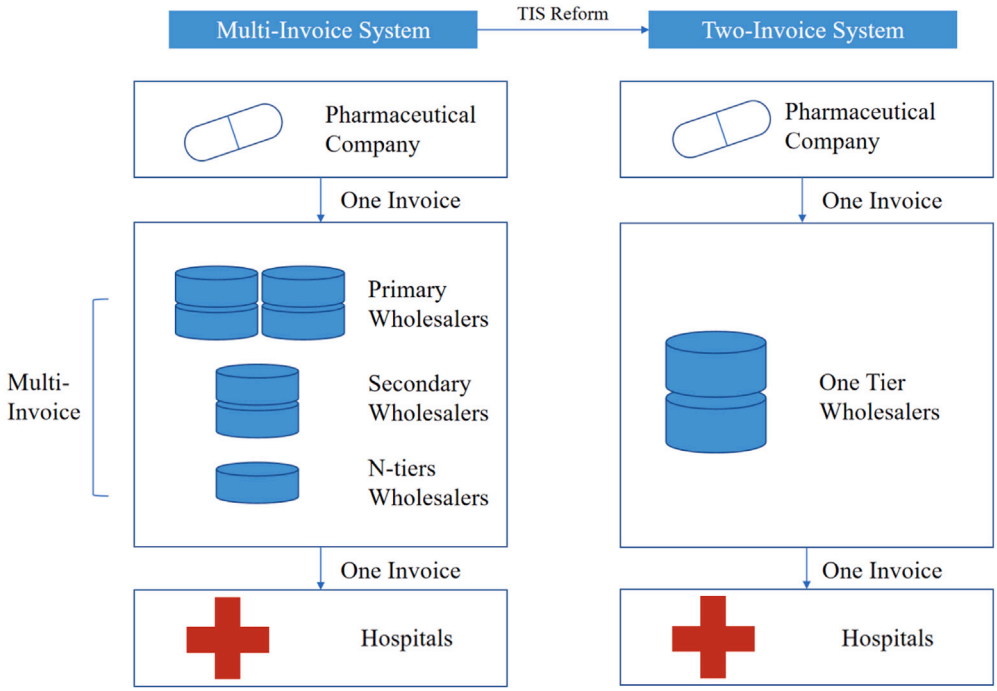

Fig. 1. Distribution of drugs before and After TIS reform.
Notes: This figure shows changes in drug distribution before and after the TIS reform. Before the reform, multiple invoices were generated across multiple tiers of distributors. After the reform, there are only two invoices: one for the flow of manufacturers to distributors, and one from distributors to end users (public hospitals).

and market dynamics remains uncertain, underscoring the need for further theoretical and empirical studies to fully evaluate the policy impact.[9]

Despite the nationwide impact and practical implications of TIS, empirical studies on the effectiveness of TIS remain limited and fragmented.[10] To investigate the effects of TIS, we first develop a theoretical model to capture the strategic responses of manufacturers and distributors. Our model indicates that streamlining the distribution chain influences pricing strategies and promotional activities in several important ways. First, whether retail prices decline as intended by the policy depends on which driving force prevails. Specifically, in supply chains with already low markups, the disruptive effects on efficiency may outweigh the benefits of eliminating double marginalization, potentially leading to an unexpected increase in retail prices. Second, overall marketing activities are likely to contract as efficiency disruptions raise marginal costs. At the same time, as manufacturers' relative efficiency may improve within the supply chain, they internalize functions that were previously outsourced, prompting an expansion of their marketing efforts.

Empirically, we employ a staggered Difference-in-Differences (DiD) approach, leveraging the phased rollout of TIS across Chinese provinces. We find a modest short-run increase in drug prices, underscoring the complexity of its market effects and highlighting trade-offs inherent in the reform. The heterogeneity analysis supports our model's prediction that supply chains with thinner margins are more vulnerable to efficiency disruptions. Specifically, drugs with lower retail prices or those sold in economically developed regions—where competition is more intense—exhibit greater price increases.

We further explore the broader economic consequences of this policy shock on China's pharmaceutical supply chain, focusing on financial data from listed pharmaceutical manufacturers compared to other healthcare firms, mainly API producers and medical device manufacturers. In particular, pharmaceutical manufacturers have significantly increased their sales and marketing expenses, suggesting possibly an expansion of their in-house distribution activities as an adaptation to regulatory disruptions. This aligns with anecdotal evidence pointing to a longstanding issue in China's pharmaceutical sector: commercial bribery. Historically, a substantial portion of drug spending was transferred to prescribing physicians. By enhancing transparency in intermediary profits, TIS aimed

[9] We set aside in our focus analysis the post-reform consolidation of intermediaries and its potential price effects, as China's TIS does not appear to substantially enhance their bargaining powers. Although the policy expelled some small distributors by mandating two invoices in sales to public hospitals, it does not exclude other sales channels. Industry concentration has indeed risen; however, the market remains fragmented with over 10,000 distributors, listed firms accounting for only about 30% of revenues, and persistently low profit margins (1.7%–2.0% from year 2011 to year 2019). For more detailed discussions on this, please refer to Section 3.1.

[10] For some anecdotal evidence, see Ye and Ying (2017), Qiao et al. (2018), Zheng et al. (2018).

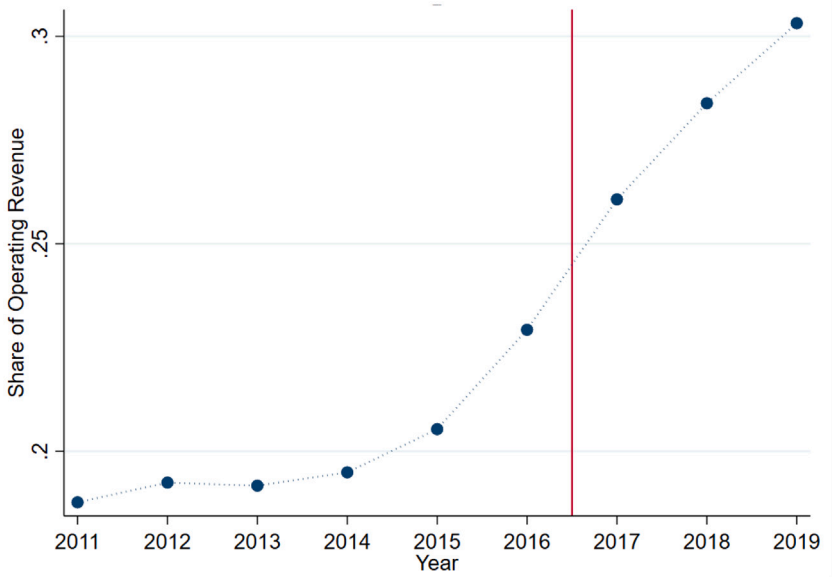

Fig. 2. Share of operating revenue for listed pharmaceutical distributors.
Notes: This figure shows the ratio of revenue of listed companies in the pharmaceutical distribution industry to the total industry revenue. The detailed information about the listed pharmaceutical distributors is provided in Appendix B.1.

to address this issue. However, our findings imply that these practices may have shifted upstream, with manufacturers absorbing these costs instead.

To the best of our knowledge, our research is the first comprehensive study to examine the national impact of TIS on drug prices. Previous studies have focused on provincial or city-level data (Ye and Ying, 2017; Qiao et al., 2018), or on case studies and interviews (Zheng et al., 2018). However, these studies provide inconclusive results. Qiao et al. (2018) found that ex-factory prices increased while retail prices decreased, whereas Zheng et al. (2018) reported that 15% of sales managers disagreed with claims of price reductions. Moreover, these studies lacked rigorous causal analysis. Our study fills this gap by offering a national-level evaluation of the causal effects of TIS on drug prices, while also exploring the conflicting mechanisms driving these outcomes.

Our study contributes to the broader health policy discourse on controlling escalating healthcare expenditures. Besides examples including generic compulsory substitution in the U.S., reference pricing in Europe (Bardey et al., 2010; Brekke et al., 2011), studies suggest that improving distribution systems could help control rising drug prices (Agwunobi and London, 2009; Schneller et al., 2023). Our findings indicate that, without properly balancing efficiency losses, administrative measures to reduce double marginalization may fail to achieve their policy objectives.

Our paper also contributes to the broader IO literature on transaction costs and vertical integration. According to pioneer studies in transaction costs theory (Williamson, 1975, 1979, 1985; Klein et al., 1978), vertical integration occurs when trading relationships become inefficient. When contracts are incomplete, and the value of internal transactions exceeds that of market exchanges, firms may internalize transactions. Our study examines how regulatory interventions like TIS can force vertical integration by reducing trading opportunities in the market. Because the regulation is exogenous, the internal efficiency trade-off is muted; integration may therefore be less efficient, as evidenced by our findings.

Finally, our paper may contribute to the ongoing policy debate about government intervention in addressing market failure. In light of the seminal work of Arrow (1963), numerous regulatory measures have been introduced in several countries to address market failures in healthcare (Anis and Wen, 1998; Wright, 2004; Dubois et al., 2021; Lauenroth et al., 2020; Brekke et al., 2023; Cao et al., 2024). In China, distributors are potentially implicated in commercial promotions and bribery in the name of sales expenses, which inflates drug prices (Yu et al., 2010; Hasan et al., 2019; Ran et al., 2022). Our analysis of TIS presents evidence of an intervention in the Chinese pharmaceutical distribution sector aimed at alleviating double marginalization and marketing activities. We suggest that mechanically altering market structures can generate unintended consequences alongside potential gains.[11]

Our paper is structured as follows. Section 2 describes the implementation of TIS in China; Section 3 proposes an illustrative theoretical model; Section 4 presents data and summary statistics; Section 5 outlines our empirical framework; Section 6 discusses the empirical findings; and Section 7 concludes.

[11] However, it is important to note that our study does not offer a comprehensive benefit-cost analysis or a definitive assessment of TIS effectiveness, as we could not capture the full range of outcomes, such as impacts on tax avoidance behaviors. Instead, our work provides a snapshot of the price effects and changes in firms' financial performance resulting from TIS.

2. The implementation of Two-invoice System

The introduction of the TIS in China arose in response to long-standing concerns about inefficiencies and widespread malpractices in the pharmaceutical distribution sector. Historically, the multi-tiered distribution chain led to significant price inflation, as drugs passed through numerous intermediaries—from national to provincial, municipal, county, and district levels—before reaching hospitals. Each layer added markups, with pharmaceutical representatives often engaging in subcontracting practices, kickbacks, and commissions. As a result, factory prices frequently diverged from provincial bidding prices by several multiples, and by the time drugs reached patients, prices had escalated by tens or even dozens of times.

The concept of the TIS first appeared in the draft "2008 Guangdong Province Sunshine Procurement Implementation Plan for Medical Institutions", released by the Guangdong Provincial Health Department in 2007. Although the plan sought public feedback, it faced resistance: 35 pharmaceutical companies submitted a joint letter urging its cancellation or postponement. Due to significant industry skepticism, the TIS was included only as a non-mandatory principle in the final 2008 version of the plan and remained unenforced at that time.

Fujian Province became the first region to pilot the TIS, issuing the Three Measures for implementing the TIS on September 30, 2009.[12] Sanming City in Fujian emerged as a prominent example of successful implementation. Facing medical insurance fund deficits in 2012, the city adopted TIS alongside other reforms, such as price-capped procurement, the "One Drug, Two Strengths" policy, and Case-Diagnosis Related Groups (C-DRG) to regulate hospital expenses and prescription dosages. These measures collectively reduced the total pharmaceutical expenditure of Sanming's 22 hospitals from 780 million yuan in 2011 to 560 million yuan in 2013—a 28.2% reduction.

The Sanming reforms served as a catalyst for national policy. On April 6, 2016, the Executive Meeting of the State Council identified inflated drug prices as a key issue in healthcare reform and proposed the nationwide promotion of the TIS to enhance price transparency in pharmaceutical distribution. In the following year, the State Council and other agencies repeatedly emphasized the TIS's importance in reducing markups and regulating supply chains.[13] On January 9, 2017, the Health Care Reform Office of National Council and eight other ministries formally issued the "Notice on the Implementation of the 'Two-Invoice System' in the Drug Procurement of Public Medical Institutions (Trial)", marking the official nationwide adoption of the TIS.[14] In subsequent years, each province, municipality, and autonomous region introduced corresponding policy documents, providing implementation plans and schedules specific to their respective jurisdictions (as shown in Appendix B.2).

The TIS caps the number of VAT invoices at two—one from the manufacturer (or an entity deemed equivalent) to a distributor, and one from the distributor to the public medical institution; intra-group transfers within a distribution group do not count towards the two, but may issue at most one invoice. Public medical institutions must reconcile invoice-goods-accounts by verifying the distributor's invoice and a stamped copy of the manufacturer's purchase invoice. The TIS limits invoices, not firms; however, because any resale between independent distributors would require a third invoice (disallowed), multi-step wholesaling is effectively precluded.

The TIS aims to enhance transparency and curb excessive pricing in the pharmaceutical supply chain by reducing opportunities for intermediaries to inflate costs. By tracking drug transactions through only two invoices, it becomes easier to monitor distribution flows, mitigate risks of counterfeit drugs, and prevent unlawful markups. However, this structural shift may also have unintended consequences, such as reducing the availability and variety of pharmaceutical products, which could impact patient care and procurement strategies.

Effective implementation of the TIS requires robust government oversight. Inadequate enforcement may undermine its goals, allowing malpractices to persist. Early data from China's Ministry of Commerce indicate that the policy has had a significant impact on the distribution industry. Since 2016, the TIS's two-invoice cap effectively curtailed resale between independent distributors (any resale would require a third invoice), increasing direct sales from manufacturers to public hospitals and widening the gap between terminal and distributor sales (see Fig. 3). This trend reflects both the reduced role of intermediaries and the effective enforcement of TIS.

3. Theoretical model

3.1. Introduction of the model

In this section, we build a multiple-layer supply chain model to study the effects of TIS on the retail prices of the pharmaceutical distribution chains. The introduction of TIS effectively shortens the supply chains, reducing them to three key participants:

[12] Three Measures include "Measures for Qualification Certification and Management of Pharmaceutical Manufacturing Enterprises for Centralized Drug Procurement", "Measures for Qualification Certification and Management of Pharmaceutical Distribution Enterprises for Centralized Drug Procurement", and the "Supervision and Management Measures for Pharmaceutical Distribution in Centralized Drug Procurement".

[13] See, for example: "Notice on the Issuance of the Key Work Tasks for Deepening the Reform of the Medical and Health System in 2016.", General Office of the National Coucil, April 26, 2016, Link: https://www.gov.cn/zhengce/content/2016-04/26/content_5068131.htm (Accessed July 5, 2023); "Notice on Carrying Out the Comprehensive Reform of County-level Public Hospitals in 2016", NHFPC, September 18, 2016, Link: https://www.nhc.gov.cn/tigs/c100053/201609/66c2fbcb8b8442d2b02e4f0426efa68d.shtml (Accessed December 29, 2025); and other related documents.

[14] For the full text, see "Notice on Issuing the Implementation Opinions (Trial) on Implementing the 'Two-Invoice System' in the Drug Procurement of Public Medical Institutions". Health Care Reform Office, National Council of the People's Republic of China, January 9, 2017. Link: https://www.nhc.gov.cn/tigs/c100052/201701/9cf19f53d58946abac5658bf3669ea45.shtml (Accessed December 29, 2025).

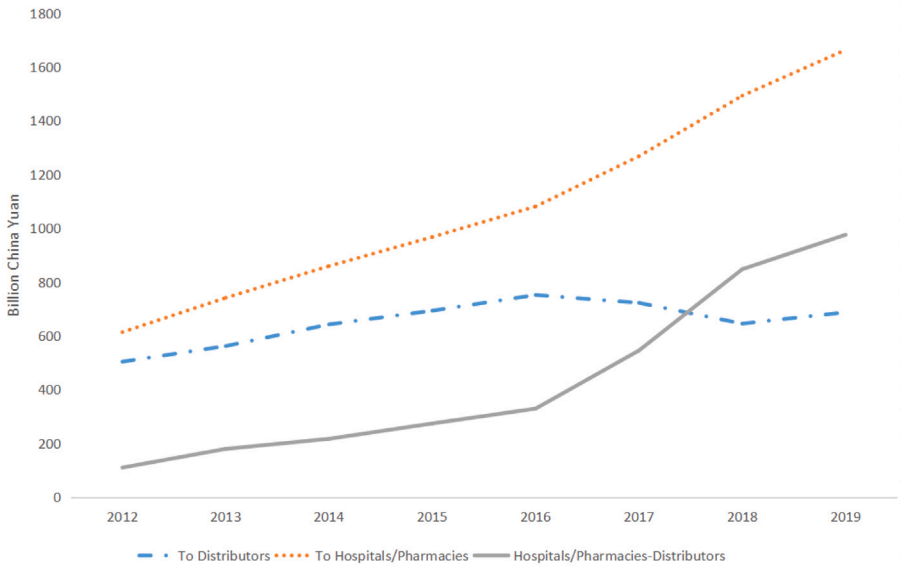

Fig. 3. Revenue flows along the pharmaceutical distribution chain: Distributors vs. Hospitals/Pharmacies and gap (2012–2019).
Notes: This figure plots pharmaceutical sales revenue along the distribution chain flowing to distributors (blue, dashed) and to hospitals/pharmacies (orange, dotted), along with their gap (grey, solid = hospitals/pharmacies - distributors). Distributor revenue levels off through 2016 and declines thereafter, while hospital/pharmacy revenue rises steadily, causing the gap to widen sharply from 2017 onward. This pattern is consistent with the Two-Invoice System (TIS) curtailing inter-distributor transactions and reducing revenue captured by intermediaries. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
Data source: Statistical Report on Pharmaceutical Circulation Industry, 2012–2019, Ministry of Commerce, PRC. Available at: https://yplt.mofcom.gov.cn/stat/page/auth/DrugWall.html.

manufacturers, distributors, and consumers/public hospitals, helps eliminate double marginalization but may also decrease efficiency across the entire supply chain.[15,16] We solve the model to explore the firms' decisions regarding promotion and pricing within this vertical supply chain structure, and examine how these equilibrium variables respond to the change in supply chain length. The aim is to derive insights into the policy effects of TIS on drug prices.

It is noteworthy that, since the policy administratively limits the chains to public hospitals to a single distributor, small and local distributors focusing on this channel may be expelled from the market, and it is natural to expect consolidation—large, more efficient distributors acquiring smaller ones—thereby increasing downstream market power and potentially influencing both the ex-factory and retail prices. We do not incorporate such an effect into our pricing model of a multi-tier vertical distribution structure, as it would substantially distract from our primary focus. Meanwhile, distributors in China's pharmaceutical market remain competitive even after the TIS reform (more than 10,000 distributors), with no significant change in industry profit margins,[17] and the market is not dominated by a small number of giants (the ratio of revenue of listed companies in the pharmaceutical distribution industry to the total industry revenue has largely increased after TIS, but is still just around 30%). We believe the impact of TIS on distributors' market power is limited, at least in the time window of our observations. Therefore, our paper focuses primarily on the impact of administratively shortening the supply chain on efficiency and double marginalization.[18,19]

[15] We adopt a multi-layer wholesale pricing model, for which it is well recognized that double marginalization arises as each intermediary in the distribution chain imposes its own markup, resulting in cumulative price distortions and reduced efficiency (Spengler, 1950; Tirole, 1988).

[16] In the simple one-manufacturer–one-retailer settings, double marginalization can be eliminated with complete and enforceable contracts (Tirole, 1988), such as two-part tariffs, revenue-sharing, or resale price maintenance (not naive as the wholesale ones). In practice, however, frictions such as information asymmetries, contractual incompleteness, unobservable efforts, and legal restrictions often hinder such coordination (Bolton and Whinston, 1993; Cachon, 2003; Lafontaine and Slade, 2008). As discussed in Dong et al. (1999) and Economic Research Institute Task Force, National Development and Reform Commission et al. (2014), China's pharmaceutical supply chains historically evolved into a multi-tiered "province-city-county" structure, and due to local protectionism, these tiers are not easily dismantled or adjusted by market forces.

[17] According to the Statistical Analysis Report on the Operation of China's Pharmaceutical Distribution Industry from 2011 to 2019 issued by the Ministry of Commerce, the profit margins for the pharmaceutical distribution industry were 2.0%, 1.9%, 1.7%, 1.7%, 1.7%, 1.7%, 1.8%, 1.7%, 1.9%, and 1.7% respectively.

[18] Although our model does not capture the market power of distributors, fortunately, there is a large literature on how increased intermediary market power affects wholesale and retail prices. Intermediaries can acquire buyer (countervailing) power through mergers, which (in various forms) often reduce wholesale prices (Galbraith, 1954; Chipty and Snyder, 1999; Li et al., 2014; Loertscher and Marx, 2019), and may shift surplus away from manufacturers (Comanor and Rey, 2000). The implications for the retail prices are context-dependent—improved efficiency and cost savings can lower prices in some settings, but pass-through can be limited in others (von Ungern-Sternberg, 1996; Dobson and Waterson, 1997; Inderst and Valletti, 2011; Hosken et al., 2018; Rickert et al., 2021). Gaudin

Our model assumes a sequential pricing structure. Specifically, each firm observes the wholesale price set by the upstream firm and subsequently determines its own wholesale price for the downstream firm in the supply chain. The final price, the retail price paid by consumers (hospitals), is set at the downstream end. This pricing structure, combined with the demand function, determines the amounts of drugs produced, distributed, and consumed throughout the chain. This sequential decision-making process forms a dynamic game in which each firm optimizes its pricing based on the unit purchasing cost from its immediate upstream supplier. The game is solved backward, starting from the retail price and working upward through the supply chain.

In addition to the purchasing costs, each firm incurs marketing expenses associated with promotional activities. In our model, we assume a demand expansion effect by marketing, so the optimal amount of total promotional activities equates marginal benefit and cost. We take a cooperative approach to study how this marketing budget is distributed among firms in the supply chain and find that it depends on their respective marketing efficiencies, which are influenced by their access to local information and social networks. There is an inherent conflict of interest in allocating the promotional budget. Promotional activities—which may include legitimate efforts such as advertising and academic sponsorship, as well as illicit practices such as providing kickbacks intended to influence hospital procurement or physicians' prescribing—tend to generate spillover benefits for all firms operating along the same supply chain by boosting overall drug sales. However, the financial costs of these activities, along with the associated legal and regulatory risks in the case of potentially unlawful conduct, are borne exclusively by the firms that undertake them. Consequently, firms must make a joint decision regarding how to share the promotion expenses, a decision that is influenced by their relative bargaining power.[20]

We show that this allocation decision can be simplified if made at the outset of the game, with the outcome being that promotion activities are allocated to the firms with the lowest marketing cost. This result arises because firms internalize the promotional costs by adjusting their wholesale prices accordingly as the game progresses along the supply chain.

Our model predicts two main outcomes related to the introduction of the TIS. The first prediction concerns the shift of promotional activities upstream to the manufacturer. Under a long supply chain, local distributors often bear the brunt of promotional expenses due to their proximity to consumers and better local market knowledge. However, with the introduction of TIS, vertical integration occurs, and distributors with local information and social networks are consolidated into larger, fewer entities. This consolidation may erode the marketing efficiency associated with local distributors, as much of the local knowledge and networks could be lost. Furthermore, large, integrated distributors are now subject to closer regulatory scrutiny, leading to a higher expected regulatory risk compared to smaller, local distributors in the previous longer supply chain. As a result, manufacturers are likely to absorb more marketing costs post-TIS, having their marketing efficiency relatively increased. Our empirical results, presented in Section 6.7, support this prediction.

The second key prediction relates to the impact of TIS on retail prices, which is the central focus of our empirical analysis. After solving the supply chain's pricing game, we express the retail price as a function of the supply chain length. Interestingly, the relationship is not monotonic and is dependent on several factors of the supply chain, such as its markups, efficiency disruption, and sensitivity of the consumer demand to marketing activities. A shorter supply chain reduces the accumulation of profit markups along the chain, which would tend to reduce the retail price inflated by double marginalization. However, as noted earlier, vertical integration may undermine the overall marketing efficiency of the supply chain, creating upward pressure on retail prices. The interplay between these two forces—one pushing prices down and the other pushing them up—becomes an empirical question. The main contribution of our empirical work is to design a DiD test to assess whether TIS has achieved its intended policy goal on retail prices.

3.2. Main model

3.2.1. Supply chain model and results

We consider a supply chain with m layers. C_1 represents the firm that directly sells drugs to end patients or hospitals, while C_m is the top supplier (i.e., manufacturer). In this model, C_j sets a wholesale price $p_j^{(m)}$ to C_{j-1}, for $j = 2, \ldots, m$, where the superscript (m) denotes the chain's whole length. The supply chain game proceeds sequentially, with firms $C_m, \ldots, C_1$ setting prices $p_m^{(m)}, \ldots, p_1^{(m)}$, such that $p_1^{(m)}$ represents the retail price set for consumers (hospitals).

The unit production cost by the manufacturer C_m is denoted by c. The demand is characterized by the function $Q = D(\alpha) - p_1^{(m)}$, where $D(\alpha)$ is the maximum potential demand and $p_1^{(m)}$ is the retail price. α is the total amount of marketing activities carried out by

(2018) emphasizes that countervailing power translates into lower retail prices primarily under a specific demand curve (high convexity), underscoring why retail effects are theoretically ambiguous.

[19] Healthcare related evidence indicates that greater downstream buyer concentration can discipline upstream prices, while effects on retail/patient prices hinge on pass-through and market conduct (Barrette et al., 2022). Buyer size alone rarely secures wholesale discounts-these primarily arise when suppliers face competition (Ellison and Snyder, 2010). In three-tier settings with PBMs, rebate-based models and high concentration can raise list/ex-factory prices and weaken pass-through to patients (Mattingly et al., 2023).

[20] In supply chain models, many forms of marketing effort—such as local advertising, promotional activities, and retailer service investments—are difficult to contract upon, which impedes incentive alignment and coordination (Cachon, 2003). Our analysis does not incorporate non-price contractual terms beyond marketing efforts, as classic arrangements such as regional exclusivity or logistics outsourcing are either not central to the focus of this study or not directly related to the length of pharmaceutical supply chains. Contractual mechanisms governing marketing efforts, such as franchise fees and resale price maintenance (Mukhopadhyay et al., 2009) or cooperative advertising schemes (Jørgensen and Zaccour, 2014), which are more prevalent in pharmaceutical markets, generally predict a decline in marketing effort as efficiency diminishes. This pattern is consistent with the implications of our model (Corollary 5). In addition, this strand of literature does not explicitly model the policy shock that mandates a shortening of the supply chain.

the supply chain.[21] Marketing has a demand-expansion effect, so we assume $D'(\alpha) > 0$ and $D''(\alpha) < 0$. The quantity demanded, Q, is passed back through the supply chain to the upstream firms, with each firm receiving its respective wholesale price. Specifically, $p_j^{(m)}$ represents the unit purchasing cost for firm C_{j-1}, for $j = 2...m$, and we define $p_{m+1}^{(m)} = c$ for convenience.

In addition to the purchasing cost, each firm also incurs some marketing expenses. $\alpha = \sum_{j=1}^{m} \alpha_j$ so $\alpha_j \geq 0$ is the amount of marketing activities carried out by firm C_j. Assuming that R_j is the unit marketing cost for firm C_j, then firm C_j bears the marketing expense $R_j\alpha_j$ for each quantity of drugs being sold to end-point consumers.

The demand quantity $Q^{(j-1)}$ is the demand faced by firm C_j, assuming that all prior firms in the chain (from C_1 to C_{j-1}) have optimized their pricing and marketing decisions, and considering that $p_{j+1}^{(m)}$ is exogenous. The optimization problem for firm C_j is given by

$$\pi_j^{(m)} = \max_{p_j^{(m)}} Q^{(j-1)}\left(p_j^{(m)} - p_{j+1}^{(m)} - R_j\alpha_j\right).$$

Marketing activities $\{\alpha_j\}$ are assumed to be fixed for the moment, and we shall reexamine the optimization problems over $\{\alpha_j\}$ after the optimization of $\{\pi_j^{(m)}\}$ and show that all firms along the supply chain have the same objective in determining the optimal $\{\alpha_j\}$.

This problem is solved sequentially, starting from the final firm C_m and working backward to C_1. The solution to the optimization process is described by the following lemma:[22]

Lemma 1. *For $j = 1, \ldots, m$, the demand faced by firm C_j is*

$$Q^{(j-1)} = \frac{D(\alpha) - p_j^{(m)} - \sum_{i=1}^{j-1} R_i\alpha_i}{2^{(j-1)}}$$

and the optimal price set by firm C_j is

$$p_j^{(m)} = \frac{D(\alpha) - \sum_{i=1}^{j-1} R_i\alpha_i + R_j\alpha_j + p_{j+1}^{(m)}}{2}.$$

3.2.2. Retail price and marketing efficiency

Using the recursive solution of the supply chain game, we can express the retailing price as follows:

Proposition 2. *The retail price $p_1^{(m)}$ is given by*

$$p_1^{(m)}(\alpha) = \frac{(2^m - 1)D(\alpha) + c + \sum_{j=1}^{m} R_j\alpha_j}{2^m}.$$

The TIS aims to reduce the supply chain length from m layers to 2. The main policy concern is that profit margins extracted by firms at multiple layers of the supply chain may contribute to higher retail prices. By cutting the supply chain length, the TIS reduces the price inflation caused by double marginalization. On the other hand, restricting the supply chain length cannot avoid crowding out those firms that are operationally efficient. For example, marketing activities encompass not only advertising but also promotions within hospitals, and distributors located closer to consumers (hospitals) may have a comparative advantage in executing these activities, but such distributors may no longer exist after the TIS.

The supply chain firms usually negotiate to allocate the marketing activities. We assume that the marketing activities $\{\alpha_j\}$ are not set sequentially; rather, they are determined in a cooperative way, and we provide the following cooperative result from an efficiency perspective.

Recall that R_j is the unit marketing cost for firm C_j. Let $R_{\min} = \min\{R_j | 1 \leq j \leq m\}$, and $M = \{j | R_j = R_{\min}\}$ is the subset of firms whose marketing efficiencies are the highest among all. Denote $\alpha^* = \sum_{j=1}^{m} \alpha_j^*$.

Proposition 3 (*Efficiency Matters*). *Regardless of how the supply-chain firms bargain to determine the marketing activities at the outset of the sequential pricing game, it is always optimal to set $\{\alpha_j^*\}$: $D'(\sum_{j\in M} \alpha_j^*) = R_{\min}$ and $\alpha_j^* = 0$ for $j \notin M$. Moreover, if the supply-chain firms choose their marketing efforts $\{\alpha_j\}$ at the same time (in a non-cooperative way), the above cooperative solution also constitutes a Nash equilibrium.*

This result implies that when determining and allocating marketing activities among supply chain firms, it is in the interest of all firms for the most efficient firms (with the lowest R_j) to bear the entire marketing cost and to set the total marketing effort at the level that equates marginal benefit and cost. Such an optimal solution is also stable in the sense that no one has an incentive to deviate unilaterally.

[21] Within the supply chain, distributors' activities comprise passive functions such as logistics (including transportation, order fulfillment, inventory turnover, etc.), which scale with the volume of drugs sold, and active functions such as marketing and promotion, which can expand demand. In China, logistics services are highly competitive and, in pharmaceuticals, relatively low-cost (Alicke and Lösch, 2010), except for certain biopharmaceuticals and injectables. The principal force inflating drug prices has historically been promotion, including commercial bribery. Accordingly, our analysis focuses on the trade-off between eliminating double marginalization and preserving marketing efficiency.

[22] Proofs of results in this section are put in Appendix A, including that for Lemma 1, Proposition 2–3.

3.2.3. Effect of TIS on retail prices

We now turn to the policy implications of the TIS. Specifically, we consider how the shortening of the supply chain under the TIS impacts the retail price set for hospitals. Proposition 2 characterizes the retail price as a function of the marketing efforts $\{\alpha_j\}$. Proposition 3 implies that an optimal solution can be found by allocating all marketing activities for a sole firm with the lowest unit selling cost along the chain and having it equalize marginal benefit and cost. We put the subscript m on α_j^* and α^* to emphasize the optimal solution for a supply chain of length m: $\alpha^{*,m} = \sum_{j=1}^{m} \alpha_j^{*,m}$. We obtain the retail price before and after the implementation of the TIS as follows:

Proposition 4. *When the supply chain length is reduced, the effect on the retail price is ambiguous. Before the TIS, the retail price is*

$$p_1^{(m)} = \frac{(2^m - 1)D(\alpha^{*,m}) + c + \min_{1\le j\le m} R_j}{2^m};$$

After the TIS, when the supply chain is reduced to two layers, the retail price is

$$p_1^{(2)} = \frac{3D(\alpha^{*,2}) + c + \min(\tilde{R}_1, \tilde{R}_2)}{4}.$$

Recall that C_j is the distributor with j layers to end consumer. We use $\tilde{R}_1$ and $\tilde{R}_2$ to demonstrate that, after TIS, the positions of C_1 and C_2 may have been replaced by two other players in the supply chain: a surviving distributor ($\tilde{C}_1$), and the manufacturer ($\tilde{C}_2$), respectively, thus $\tilde{C}_2 = C_m$ and $\tilde{C}_1 \in \{C_1, \dots, C_{m-1}\}$. In general: $\min(\tilde{R}_1, \tilde{R}_2) > \min_{1\le j\le m} R_j$.

Following Proposition 4, we make the following observation concerning the TIS's policy effect on drugs' retail prices.

Observation

$$\begin{aligned} & p_1^{(2)} - p_1^{(m)} \\ = & \left(1 - \frac{1}{2^m}\right)\left[D(\alpha^{*,2}) - D(\alpha^{*,m})\right] - \left(\frac{1}{4} - \frac{1}{2^m}\right)\left[D(\alpha^{*,2}) - c - \min(\tilde{R}_1, \tilde{R}_2)\right] \\ & + \frac{1}{2^m}\left[\min(\tilde{R}_1, \tilde{R}_2) - \min_{1\le j\le m} R_j\right], \end{aligned}$$

such that the change in retail price is dependent on three terms:

- Demand deflation by marketing reduction: $(1 - \frac{1}{2^m})\left[D(\alpha^{*,2}) - D(\alpha^{*,m})\right] < 0$;
- Elimination of double marginalization (markups): $-\left(\frac{1}{4} - \frac{1}{2^m}\right)\left[D(\alpha^{*,2}) - c - \min(\tilde{R}_1, \tilde{R}_2)\right] < 0$;
- Efficiency disruption: $\frac{1}{2^m}\left[\min(\tilde{R}_1, \tilde{R}_2) - \min_{1\le j\le m} R_j\right] > 0$.

The implementation of TIS shortens the supply chain from m layers to 2 layers, which reduces the price inflation caused by double marginalization. This effect is significant when the markups are high along the chain, that is, when the market's fundamental demand significantly surpasses the cost and expense items.

However, retaining only two layers may not necessarily lead to a lower drug price, especially when the supply chain efficiency is largely disrupted. For example, if smaller distributors with long-standing relationships with hospitals are merged or eliminated and replaced by larger distributors, overall efficiency can decrease and the retail price increases $\left(p_1^{(2)} > p_1^{(m)}\right)$. This is the case when the difference between $\min(\tilde{R}_1, \tilde{R}_2)$ and $\min_{1\le j\le m} R_j$ is large, so there is a direct effect on price increase through marketing expense.

Nevertheless, there is also an indirect effect through the demand expansion effect by marketing, which decreases the price. Since $D(\cdot)$ is increasing and concave in α, we have $D'(\alpha^{*,m}) = \min_{1\le j\le m} R_j < \min\{\tilde{R}_1, \tilde{R}_2\} = D'(\alpha^{*,2})$, so $\alpha^{*,2} < \alpha^{*,m}$ and $D(\alpha^{*,2}) < D(\alpha^{*,m})$. This effect is larger when the fundamental demand D is more sensitive to marketing activity α. We state this result by the following corollary.

Corollary 5. *The overall marketing efforts of the supply chain shrink after TIS, that is, $\alpha^{*,2} < \alpha^{*,m}$.*

Optimal length of a supply chain Proposition 4 and the subsequent analysis focus on the retail price effect of TIS. The theoretical prediction indicates that, while eliminating the double marginalization problem, TIS might have introduced efficiency disruption to the existing multi-layer distribution system. This implies that the mandatory administrative shortening of the supply chain to a single middleman may not be socially optimal. Leveraging the theoretical model and results, we provide views on the following normative question: namely, what would be the optimal length of a supply chain? Let $t \in \{2, \dots, m\}$ be the length of a supply chain, the price difference after a t-invoice system is then

$$\begin{aligned} & p_1^{(t)} - p_1^{(m)} \\ = & \left(1 - \frac{1}{2^m}\right)\left[D(\alpha^{*,t}) - D(\alpha^{*,m})\right] - \left(\frac{1}{2^t} - \frac{1}{2^m}\right)\left[D(\alpha^{*,t}) - c - \min_{1\le j\le t} \tilde{R}_j\right] + \frac{1}{2^m}\left[\min_{1\le j\le t} \tilde{R}_j - \min_{1\le j\le m} R_j\right]. \end{aligned}$$

We fix m and see how can we set t to minimize $p_1^{(t)} - p_1^{(m)}$ (maximize the price reduction). A smaller t implies a larger efficiency disruption since $\min_{1\le j\le t} \tilde{R}_j$ is larger. The consequence is that the markup per layer $(\frac{1}{2^t} - \frac{1}{2^m})\left(D(\alpha^{*,t}) - c - \min_{1\le j\le t} \tilde{R}_j\right)$ is smaller, since $D'(\alpha^{*,t}) = \min_{1\le j\le t} \tilde{R}_j$ is also large, and both $\alpha^{*,t}$ and $D(\alpha^{*,t})$ are small. The optimal design of the length t should balance the changes in efficiency disruption and elimination of double marginalization, and take the supply chain's characteristics into consideration.

For example, for supply chains that have small markups (market competition is intense or profit margins are low) and efficient distributors, executing the TIS policy to the letter might mean a big efficiency loss.

Our model shows that how TIS impacts the retail prices depends on the competitive outcome of two opposite forces, the elimination of double marginalization that reduces the price, versus efficiency disruption that pushes up the price. Reduction in the number of supply chain layers does not guarantee a lower retail price, as it can disrupt the efficiency of marketing activities. The overall effect of TIS on drug prices is thus an empirical question that requires careful analysis, as discussed in the empirical section of the paper. For the ease of illustration, we form a hypothesis for empirical analysis:

Hypothesis 1. For those supply chains whose markups are limited and the demands are not that sensitive to prices, the efficiency disruption effect dominates that from the elimination of double marginalization, and the drug prices may increase after TIS.

Our analysis on the optimal marketing activities revealed that $\alpha^{*,2} < \alpha^{*,m}$, so after the TIS, the supply chain exerts less overall effort on marketing since there is an efficiency disruption due to the mandatory policy. Moreover, it is in the supply chain's interest to allocate the marketing activities to those firms that are most efficient. Before TIS, distributors close to hospitals have local connections who are efficient in promoting drugs. With TIS, small distributors are expelled from the market or are merged. With two invoices only, the supply chain survives the manufacturer and one distributor. Remaining distributors are usually large and are now under closer examination. Considering that in China, commercial promotion in the drug industry are usually associated with illegal rebates and even commercial bribery, the expected marketing cost (risk) for remaining distributors shall increase. When there were many small and local distributors, manufacturers had no interest in promotion (besides the usual commercial advertisements); after the TIS, they were somehow forced to undertake a considerable amount of marketing activities.

Hypothesis 2. The overall marketing efforts conducted by manufacturers are expected to increase after TIS.

4. Data

The data used in this study to analyze the impact of TIS on drug prices and promotional activities is sourced from multiple databases, with primary data on drug procurement prices purchased from a leading pharmaceutical industry consulting firm. Provincial drug procurement prices, which are typically published in official documents or on online trading platforms, were collected by the firm since the implementation of drug procurement auctions in various provinces. Financial data of listed companies were obtained from the WIND database. Additionally, we sourced TIS implementation information from provincial health commission websites and province-level economic and demographic characteristics from the National Bureau of Statistics of China.

The drug procurement price data is structured at the product-province-month level, covering the period from January 2015 to December 2019. This dataset includes key variables such as drug procurement prices, producer name, procurement time, and procurement province. These prices reflect the amount at which public hospitals purchase drugs directly from distributors or pharmaceutical companies. Under China's new healthcare reform (since 2009), the "province-based, government-led" centralized drug bidding and procurement policy mandates that all drug procurement transactions between public hospitals and pharmaceutical firms adhere to the listed prices on the provincial procurement platforms.[23] The data provides comprehensive information on procurement prices, including details about the active ingredient, formulation, and strength of the drugs.

4.1. Dependent variable: Drug price

The dependent variable in our primary analysis is the drug procurement price. We process the data as follows: First, we treat drugs with different strengths as different products, as they often target different patient groups and demand characteristics. For example, a 5-mg finasteride primarily treats prostatitis, while a 1-mg tablet addresses hair loss. Second, the original dataset only includes updated prices for each product listed on the provincial procurement platform. If a product's price does not change in a given month, it is not recorded. To construct the panel dataset, we carry forward the most recent procurement price when no new price record is available for a particular month. Finally, we calculate the dependent variable as the average unit price (per tablet or pill) at the province-month-company-active ingredient-strength level and log-transform this price to account for potential skewness.

For example, if in January 2016, two pharmaceutical companies—company A and company B—win the auction for a box of 10 aspirin tablets (10 mg each) in Beijing, with bid prices of 100 Yuan and 110 Yuan, respectively, the drug price for "Beijing-January 2016-Aspirin-10mg/Pill" would be the average of these bids, i.e., 10.5 Yuan per pill. All price data are adjusted to 2016 Yuan using the Consumer Price Index (CPI). To mitigate the impact of extreme values, we exclude the top and bottom 5% of the most expensive and least expensive drugs from the main analysis.[24]

[23] Our analysis focuses primarily on the drug distribution pathway from manufacturers to distributors and, ultimately, to hospitals, which accounts for over 77% of the drug distribution in the Chinese pharmaceutical market (RolandBerger, 2015). We exclude the distribution within hospitals, as the price margin from hospitals to patients is regulated under the zero-markup drug policy.

[24] We use the full sample in a robustness check. Results are shown in Appendix B.3.

Table 1
Definition of variables.

Dependent variables	Definitions	Sources
Price per Pill	Average unit prices at ingredient-firm-strength level	Yaozh.com (2015–2019)
Sales and Marketing Expenses	Costs incurred by the sales department	WIND (2011–2019)
R & D Expenses	Costs associated with research and development	WIND (2011–2019)
Procurement Distance	Driving distance from pharmaceutical companies to provincial capitals	Manual collection using online map
Independent Variables	**Definitions**	**Sources**
Population	Provincial population	NBSC (2015–2019)
GDP	Provincial GDP	NBSC (2015–2019)
Elderly Dependency Ratio	Ratio of the number of older persons (>65) to the number of persons of working age	NBSC (2015–2019)
Num of Hospitals	Number of hospitals	NBSC (2015–2019)
Num of Physicians	Number of registered physicians	NBSC (2015–2019)
Num of Beds	Number of hospital beds	NBSC (2015–2019)
Gov Healthcare Exp	Goverment expenditure on healthcare sector	NBSC (2015–2019)
Social Health Insur Exp	Social health insurance expenditure	NBSC (2015–2019)
Pharmaceutical Industry Output	Total revenue from pharmaceutical companies	WIND (2011–2019)
Revenue	Total revenue from a business's primary operations	WIND (2011–2019)
ROA	Return on assets	WIND (2011–2019)
Assets	Total assets reported on the balance sheet	WIND (2011–2019)
ROE	Return on equity	WIND (2011–2019)
Leverage Ratio	Debt/Assets	WIND (2011–2019)
Shareholder	Shareholding of top five shareholders	WIND (2011–2019)
Seller	Number of employees engaged in marketing	WIND (2011–2019)

Notes: This table shows the definitions of the main variables used in this study and the data sources. NBSC stands for the National Bureau of Statistics of China. WIND is the leading financial database in China.

4.2. Sample selection

For the main analysis, we focus on tablets and capsules, as these drug formulations are more standardized compared to other forms such as suppositories or creams. We exclude traditional Chinese medicines and drug combinations, concentrating solely on individual active ingredients. The final sample includes 241 active ingredients, 3805 ingredient-province-strength-formulation units, and 149,013 observations.

Given that the TIS restricts the number of distributors in the supply chain, potentially influencing the range of pharmaceutical companies available to provinces, we also calculate the driving distances from the bid-winning pharmaceutical firms to the provincial capitals using Amap. This allows us to assess whether the TIS impacts the distribution distances for drugs procured in each province.

4.3. Financial data

To explore the impact of the TIS on the financial performance of pharmaceutical firms, we obtain financial data for listed pharmaceutical manufacturers and other health-related firms from the WIND database. This database provides comprehensive firm-specific financial data for A-share listed companies. We focus on four main types of listed firms: pharmaceutical manufacturers (our treatment group), Active Pharmaceutical Ingredient (API) manufacturers, pharmaceutical intermediates manufacturers, and medical device firms (the control group). We argue that while TIS mainly affects the distribution chains within the pharmaceutical sector, API manufacturers, pharmaceutical intermediates manufacturers, and medical device firms are less likely to be directly influenced by the policy, though they share similar market trends with pharmaceutical manufacturers.

The financial data includes information on 304 publicly listed pharmaceutical manufacturing firms from 2011 to 2019. Key variables include total revenue, leverage, sales and marketing expenses (including advertising, promotion, and selling expenses), R&D expenses, operating profit margin, firm size, year of establishment, majority shareholder ownership, return on equity, company type, location, and return on assets. Additionally, we incorporate selling and R&D expense intensities (i.e., expense/assets) to control for firm size effects. Table 1 provides a detailed list of all variables used in the analysis, including their definitions and data sources.

4.4. Policy and demographic data

To exploit regional and temporal variations in the policy's impact, we manually gathered detailed information on TIS implementation from the websites of provincial health commissions. This includes the dates of policy announcements, the initiation dates of TIS, the duration of transition periods (used to calculate the dates of full TIS implementation), and the regions or hospitals selected for trial implementations.

We also incorporate provincial-level demographic and economic variables to control for demand-side effects on drug prices. These include annual provincial GDP, population, and the elderly dependency ratio—proxies for the socioeconomic conditions within each

Table 2
Summary statistics for variables.

Panel A: Drug prices and province characteristics	Before Treat	After Treat
Price per Pill	2.507	2.535**
Population (million)	47.220	48.480
GDP (trillion)	29.644	33.851***
Elderly Dependency Ratio	14.341	16.639***
Driving Distance	136.740	138.312**
Num of Hospitals	572.471	640.437***
Num of Physicians (thousand)	97.771	109.305**
Num of Beds (thousand)	250.786	287.616**
Gov Healthcare Exp (trillion RMB)	46.533	54.197***
Social Health Insur Exp (million RMB)	3554.264	4524.605***
Pharmaceutical Industry Output (trillion RMB)	2.753	3.212***
Observations	76,235	72,778
Panel B: Finished pharmaceutical manufacturers financials	**Before Treat**	**After Treat**
Sales and Marketing Expenses (million RMB)	374.983	746.761***
R&D Expenses (million RMB)	67.547	158.135***
Assets (million RMB)	3128.220	4128.731***
ROA (%)	8.319	8.380***
Leverage (%)	0.339	0.335
ROE (%)	5.686	9.250***
Shareholder	55.633	53.437
Seller	387.651	1260.872***
Observations	552	907

Notes: In 2017 the RMB-US$ exchange rate was approximately 7:1. All prices are adjusted by CPI to 2016.

province and the healthcare burden associated with an aging population. These factors can influence drug demand and, consequently, pricing dynamics. Additionally, we control for the annual total sales of A-share listed companies in each province, reflecting the development stage of the local pharmaceutical industry and its potential influence on drug prices, particularly through supply chain efficiencies and delivery mechanisms. We further control for variables that reflect the level of development of healthcare systems in each province, including the number of hospitals, the number of registered physicians, the number of hospital beds, the total output value of the province's pharmaceutical industry, government healthcare expenditure, and social health insurance expenditure.

Our dataset allows for a comprehensive analysis of the factors influencing drug prices in China, including the role of TIS implementation and its impact on pharmaceutical distribution and promotional activities. By leveraging both firm-level and provincial-level data, we can examine how TIS affects drug procurement prices and the broader financial performance of pharmaceutical firms, while accounting for other important variables such as demographic factors and regional economic conditions.

4.5. Descriptive statistics

Table 2 presents summary statistics for the key dependent and independent variables used in our analysis. Panel A provides an overview of the average drug price per pill, alongside provincial-level demographic and economic metrics such as population, GDP, and the elderly dependency ratio, both before and after the implementation of TIS. The table shows a slight increase in the average drug price per pill, rising from 2.50 Yuan to 2.54 Yuan following the enactment of TIS.

Concurrently, we observe significant increases in both GDP and the elderly dependency ratio across provinces, which suggests that the observed price changes may be influenced by broader demographic and economic shifts. This highlights the need for a more robust identification strategy to disentangle the specific impact of drug prices, controlling for these confounding factors.

Panel B shifts the focus to the financial performance of pharmaceutical manufacturers post-TIS implementation, presenting changes in sales and marketing expenses, R&D expenses, and other relevant financial indicators. Notably, sales and marketing expenses increased largely, reflecting a substantial adjustment in the cost structure of pharmaceutical firms. This increase may be attributed to the regulatory changes introduced by TIS, which likely prompted firms to adapt their promotional and marketing strategies in response to the new competitive environment. The financial data presented in this panel sheds light on the strategic adjustments made by pharmaceutical companies in response to TIS policy changes. All prices in the table are adjusted to 2016 Yuan using the CPI to account for inflationary effects over time.

In sum, Table 2 provides a descriptive foundation for the subsequent regression analysis presented in Table 3. It offers a comparison of key variables before and after TIS implementation, which serves as a baseline for examining the causal effects of TIS on drug pricing and the financial dynamics within the pharmaceutical sector.

5. Empirical strategy

5.1. Empirical model

To identify the policy impact of TIS on drug prices, we adopt a staggered DiD methodology, leveraging the variation in the timing of the TIS implementation across provinces. By utilizing monthly-level variations in TIS implementation and our monthly

drug price data, we begin with a two-way fixed effects (TWFE) model as a baseline framework to estimate the causal effect of TIS on drug prices.

The model is specified as:

$$DrugPrice_{ipm} = \alpha + \gamma PostTIS_{pm} + \beta X_{ipm} + \delta_{ip} + \lambda_m + \varepsilon_{ipm} \tag{1}$$

where $DrugPrice_{ipm}$ represents the natural logarithm of the price for drug i procured by province p in month m; $PostTIS_{pm}$ is an indicator variable equal to 1 if TIS has been implemented in province p in month m, and 0 otherwise. Specifically, we distinguish between two policy time points: TIS Initiation, when TIS was first implemented in a province, and TIS Completion, marking the full implementation of the TIS reform, which typically involves a transition period of three to six months.

The model includes year-month fixed effects, denoted by λ_m, to capture time-varying, province-invariant factors. The drug-specific (or disease-specific) effects, invariant over time but varying across provinces, are accounted for by the molecule-province fixed effects δ_{ip}. To account for macroeconomic factors that may influence drug demand at the provincial level, we incorporate a vector of province-level covariates, X_{ipm}, including GDP, population, elderly dependency ratio, number of hospitals, number of physicians, number of hospital beds, government expenditure on healthcare, total output value of the pharmaceutical industry, and social health insurance expenditure at the province-year level. The coefficient of interest is γ, which captures the effect of the TIS reform on drug prices. Standard errors are clustered at the province-ATC major category level, as the number of provinces in the analysis is limited to 30 (Cameron and Miller, 2015).[25]

Our identification assumption requires that, in the absence of the TIS reform, drug prices in all provinces would have followed parallel trends. To test this assumption, we conduct an event-study analysis by estimating the coefficients for each period relative to TIS implementation:

$$DrugPrice_{ipm} = \alpha + \sum_{m=-11}^{12} \gamma_m TIS_p * 1(t - t_p^* = m) + \beta X_{ipm} + \delta_{ip} + \lambda_m + \epsilon_{ipm} \tag{2}$$

where t_p^* denotes the implementation time for province p, and TIS_p is an indicator variable for whether province p is in the treatment group. If the coefficients γ_m for pre-treatment periods (i.e., $m < 0$) are not significantly different from zero, we fail to reject the parallel trend assumption.

As noted in recent literature on staggered DiD designs (Goodman-Bacon, 2021; Sun and Abraham, 2021; Gardner, 2022; Abadie et al., 2023; Borusyak et al., 2024; De Chaisemartin and d'Haultfoeuille, 2024; Callaway and Sant'Anna, 2021), the TWFE estimator may be biased when treatments are staggered across units and when treatment effects are heterogeneous across time and units. Specifically, the average treatment effect (ATE) estimated by TWFE DiD represents a weighted average of the treatment effects across all possible comparisons of units treated at different times. To address this concern, we also apply the imputation DiD method (Gardner, 2022; Borusyak et al., 2024; Callaway and Sant'Anna, 2021). This method estimates ATE by fitting TWFE regressions using only observations from not-yet-treated units and time periods. These predictions are then used to infer the potential outcome for each treated unit, which allows us to estimate treatment effects at the unit level. These unit-level estimates are aggregated to compute the overall ATE.

In addition to the price analysis, we also examine the impact of TIS on pharmaceutical manufacturers' financial performance using a canonical DiD framework. The model is specified as:

$$Y_{it} = \alpha + \gamma Treat_i \times PostTIS_t + \beta X_{it} + \delta_i + \lambda_t + \epsilon_{it} \tag{3}$$

where Y_{it} represents firm i's financial performances in year t, with key variables including sales and marketing expenses, R&D expenses, and other financial indicators; $PostTIS_t$ is an indicator variable equal to 1 if $t \geq 2017$ (the year when most provinces implemented TIS), and 0 otherwise.

The treatment group consists of pharmaceutical manufacturers, while the control group includes API manufacturers, medical information service providers, and medical equipment manufacturers.[26,27] $Treat_i$ equals 1 for the treatment group (pharmaceutical manufacturers) and 0 for the control group. X_{it} is a vector of control variables that includes return on assets (ROA), return on equity (ROE), total assets, firm type (private, state-owned, or foreign), leverage ratio, and the proportion of equity owned by the top five shareholders. We also include firm fixed effects (δ_i) and time fixed effects (λ_t) to control for unobserved heterogeneity and time-specific shocks.

5.2. Confounding factors

The primary challenge in our study is to distinguish the direct effects of the TIS on drug prices from the influences of other concurrent healthcare policies in China, as illustrated in Fig. 4. These policies include the mandatory consistency evaluation for generic drugs (bioequivalence evaluation), introduced on March 5, 2016.[28] The introduction of the national centralized volume-based drug procurement (CVBP) policy on January 1, 2019,[29] and the zero-markup drug policy (ZMDP).

[25] We exclude Taiwan, Hong Kong, and Macau from the study, as TIS was not implemented there. Fujian is excluded for its early adoption of the policy.

[26] Although the TIS reform later extended to high-value medical consumables in 2018, we focus on its impact on drug manufacturers for this period, as medical device firms were less directly affected during our observation window.

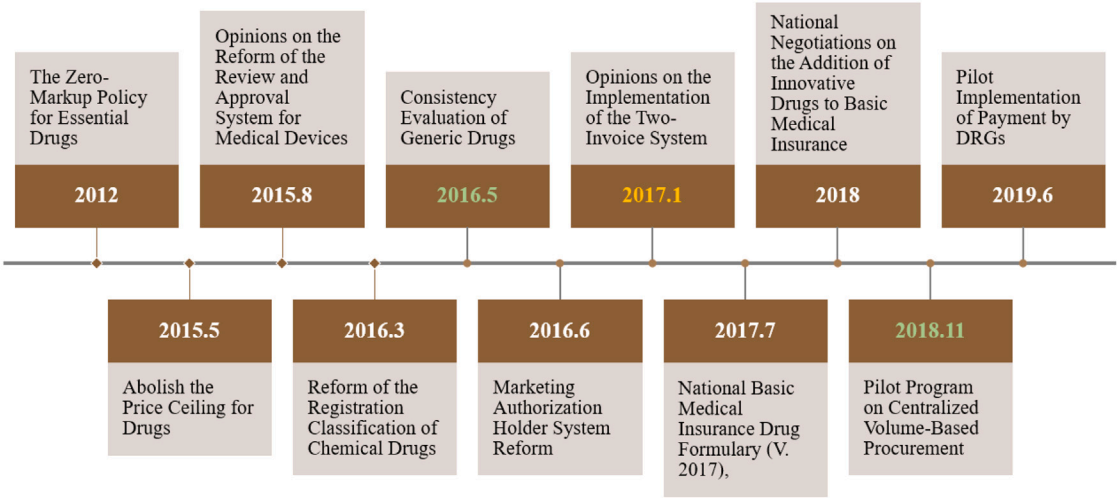

Fig. 4. Healthcare supply side reforms timeline.
Notes: This figure shows the major policy reforms aimed at the pharmaceutical industry between 2012 and 2020.

The consistency evaluation aimed to eliminate low-quality drugs from the market. This could lead to product-composition effects, where average market prices increase as only higher-quality products remain available, replacing the more diverse and lower-quality options. However, given the slow implementation of the consistency evaluation and the government's 2018 announcement extending the evaluation deadline for certain drugs until 2021,[30] we do not expect this policy to affect our estimates significantly. To address potential concerns, we conducted a robustness check by analyzing data at the ingredient-producer-province level, focusing only on drugs that had passed the consistency evaluation—those observed both before and after the TIS. Our conclusion remains unchanged, confirming the robustness of our findings.

The second policy, CVBP, is a national procurement program where the government promises large volumes of purchase in exchange for lower drug prices, similar to reforms in the U.S. Medicare Part D drug pricing. Previous studies have shown that CVBP led to an average price reduction of 65%–70% for drugs involved in the competitive bidding process (Chen et al., 2020; Yang et al., 2022; Zhu et al., 2023; Cao et al., 2024; Liu et al., 2024; He and Yang, 2024; Jiang et al., 2025). In 2019, 11 drug molecules were included in the CVBP pilot. To address this, we excluded these drugs from our robustness tests and found that our results remained consistent with those from the full sample.

The ZMDP, implemented in 2012, mandates that public hospitals should not profit from the sale of drugs, aligning procurement prices with retail prices. However, we argue that this policy has a limited impact on our study because our empirical data comes from 2015, using procurement prices rather than retail prices, thus mitigating any potential influence.

To rigorously test the effects of TIS while controlling for these confounding factors, we performed a robustness check by defining our observation unit at the ingredient-producer-province level and focusing on products that had undergone the consistency evaluation both before and after TIS implementation. This approach allowed us to isolate and examine the pure price effects attributable to TIS. Despite the complex policy landscape, our analysis confirms that the observed changes in drug pricing are directly attributable to TIS, validating our findings even in the presence of potential confounding policy influences.

6. Results

6.1. Baseline: The impact of TIS on drug prices

Table 3 presents the results of the TWFE model, which examines the impact of the TIS on drug prices. Column (1) shows that the coefficient on the post-TIS initiation variable is positive and statistically significant at the 1% level, with an estimated

[27] Given that many listed firms engage in multiple business operations, we classify a firm as a pharmaceutical manufacturer if more than half of its revenue comes from formulated drugs, and as an API or medical equipment manufacturer if more than half of its revenue comes from APIs or medical equipments.

[28] The "Opinions of the State Council on Performing Consistency Evaluation of the Quality and Efficacy of Generic Drugs" was released on March 5, 2016. It requires that "*Chemical generic drugs (in oral solid dosage form) approved before October 1, 2007, in the National Essential Medicines List (2012 edition) must complete the consistency evaluation by the end of 2018, with special cases allowed until 2021. Drugs failing to meet the evaluation will not be re-registered.*" For more details, see: https://www.gov.cn/zhengce/content/2016-03/05/content_5049364.htm (accessed November 22, 2023).

[29] Details on the CVBP pilot program can be found at: https://www.gov.cn/gongbao/content/2019/content_5361793.htm (accessed November 22, 2023).

[30] In 2018 and earlier, very few drugs passed the consistency evaluation. Specific data are reported in Appendix B.4 and available at: https://www.nmpa.gov.cn/xxgk/zhcjd/zhcjdyp/20181228161601758.html?type=pc&m=.

Table 3
Baseline: Two-way fixed effects model for the impact of TIS on drug prices.

	(1)	(2)	(3)	(4)
Post TIS Initiation	0.023***		0.019***	
	(0.005)		(0.006)	
Post TIS Completion		0.022***		0.018***
		(0.006)		(0.007)
Control Variables			Y	Y
Year-Month FE	Y	Y	Y	Y
Molecule-Province FE	Y	Y	Y	Y
Observations	149,013	149,013	149,013	149,013
R-squared	0.959	0.960	0.959	0.959

Notes: The dependent variable is the natural logarithm of CPI-adjusted drug price. Control variables include GDP, population, elderly dependency ratio, number of hospitals, number of physicians, number of beds, government expenditure on healthcare, total output value of the pharmaceutical industry, and social health insurance expenditure. Standard errors are clustered at the province by ATC major category. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

increase of 2.3% in drug prices following the reform. In Column (3), we include additional controls—GDP per capita and the elderly dependency ratio—and find that the coefficient on post-TIS initiation remains stable in both magnitude and significance, confirming the robustness of the relationship between TIS and higher drug prices.

To account for potential lags between the policy announcement and full implementation, we estimate the effect of TIS separately for the initiation and completion stages. Columns (2) and (4) show that the coefficients for both time points are similar. This is likely because the time gap between the announcement and implementation of TIS is typically short—usually less than three months—reducing the likelihood of significant lag effects that have been observed in other policy studies (Alpert, 2016). For example, in Heilongjiang province, the reform was implemented immediately with full completion by August 31, 2017,[31] and in Beijing, the reform began in March 2017 and was completed by April of the same year.[32] This short lag suggests that for our empirical analysis, the policy effects are best assessed by the initiation date.

Overall, our results indicate that the TIS reform is associated with a 1.9% increase in drug prices, as indicated by the positive and statistically significant coefficients on both the post-TIS initiation and post-TIS completion variables. This finding aligns with previous studies that have used aggregated rural—urban retail drug price indices (Peng and Zhou, 2023). While the observed price increase is modest, it suggests that administrative efforts aimed at streamlining drug distribution may not be as effective in reducing drug prices as initially expected.

6.2. Staggered DiD robustness checks and confounding factors

Table 4 presents robustness checks for the baseline TWFE estimates, utilizing alternative staggered DiD estimators and different sample specifications. The first row presents the original TWFE estimate from Model (3) of Table 3 as a benchmark. In addition to the TWFE model, the third to fifth rows show results from estimators developed by Borusyak et al. (2024), Callaway and Sant'Anna (2021), and Gardner (2022), specifically designed for staggered DiD analysis.

In the second row, we show the TWFE estimates with the characteristics of the base year. The timing of TIS reform implementation in each province was likely to be influenced by local socioeconomic and healthcare sector conditions, and ignoring these factors may lead to estimation errors. Following Hoynes and Schanzenbach (2009) and Hoynes et al. (2016) strategy, we flexibly control for differential trends in the primary determinants of TIS adoption by including interactions between base year (2016) provincial characteristics and year dummies ($Year_i * Determinants_{2016,p}$). The determinants are GDP, population, elderly dependency ratio, number of hospitals, number of physicians, number of beds, government health expenditure, total output value of the pharmaceutical industry, and social health insurance expenditure (all measured at the provincial level in 2016).

The intuition is that if these baseline characteristics help determine adoption timing, then conditioning on their 2016 values interacted with year dummies helps absorb related differential trends, strengthening the plausibility of exogenous timing. In Appendix B.5, we report a diagnostic regression of TIS implementation timing on these characteristics; the model explains over 70% of the variation ($R^2 > 0.70$). In the main text, we present the specification with $Year_i * Determinants_{2016,p}$ as a robustness check in Table 4. We do not include these interactions in every specification to avoid oversaturating the fixed effects.

As mentioned earlier, several other concurrent reforms in China's pharmaceutical sector may confound the effects of TIS.[33] To account for these confounding factors, we adjust our sample in Columns (2) and (3). In Column (2), we exclude drugs included in the "4+7 pilot centralized volume-based procurement", which substantially reduced drug prices starting in 2019. The results from the TWFE, Borusyak-Jaravel-Spiess, and Gardner estimators remain positive and statistically significant, with the TWFE estimator

[31] For more details, see: https://www.hlj.gov.cn/hlj/c108013/201710/c00_30645070.shtml (accessed April 28, 2024).

[32] Details can be found at: https://www.beijing.gov.cn/zhengce/zhengcefagui/201905/t20190522_60088.html (accessed April 28, 2024).

[33] For example, the CVBP policy, which began in 2019, significantly lowered drug prices, while the consistency evaluation regulation removed low-quality drugs from the market, potentially raising average drug prices.

Table 4
Robustness checks: alternative estimates and confounding factors.

	(1) Original sample	(2) Excluding CVBP effects	(3) Excluding consistency evaluation effects
TWFE Estimators	0.019***	0.015**	0.011***
	(0.002)	(0.006)	(0.005)
TWFE Estimator with base year characteristics	0.011**	0.012***	0.008***
	(0.004)	(0.004)	(0.002)
Borusyak-Jaravel-Spiess (2024)	0.034***	0.033***	0.009***
	(0.009)	(0.008)	(0.002)
Callaway-Santa'Anna (2021)	0.012**	0.012**	0.002
	(0.005)	(0.005)	(0.001)
Gardner (2021)	0.065***	0.081***	0.006***
	(0.015)	(0.016)	(0.002)
Observations for TWFE Estimations	149,013	139,973	238,655
Control Variables	Y	Y	Y
Year-Month FE	Y	Y	Y
Molecule-Province FE	Y	Y	
Molecule-Producer-Province FE			Y

Notes: The dependent variable for all these estimations is the natural logarithm of CPI-adjusted drug price. Columns (1)–(3) list the baseline TWFE estimators and the other three alternative robust estimators based on Borusyak et.al (2021), Callaway & Santa'Anna (2021), and Gardner (2021) for three different estimation samples. Column (1) lists the robust staggered DiD results of Table 3. Column (2) lists the results excluding 4+7 volume-based procurement. Column (3) excludes the consistency evaluation effects. The treatment time is TIS Initiation. All control variables are the same as in Table 3. Standard errors are clustered at the province by ATC major category. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

yielding a coefficient of 0.015. This suggests that after excluding the effects of CVBP, the TIS reform leads to an consistent increase in drug prices.

In Column (3), we further refine the sample by restricting it to drug products (at the molecule-producer level) that remained in the provincial market both before and after the TIS implementation. This approach rules out the composition effects and price increases resulting from the consistency evaluation regulation,[34] but it also risks excluding drugs that were removed from circulation post-TIS due to non-profitable status. Thus, estimates in Column (3) may underestimate the true effects of TIS. We anticipate that the true price effects lie between the results from Columns (2) and (3).

It is important to note that the number of observations varies across models due to the different estimation methods used. For ease of comparison, the number of observations for the most comprehensive sample in the TWFE model is presented for Columns (1) through (3). The sample in Column (3) is based on data at the molecule-producer-province-month level, which results in nearly double the number of observations compared to Column (1), which uses the molecule-province-month level.

In summary, the robustness checks using alternative DiD estimators and the exclusion of potential confounding factors confirm that the TIS reform is associated with a statistically significant increase in drug prices. Despite variations in magnitude, all methods consistently point to the same conclusion: the TIS policy, rather than reducing drug prices, is associated with higher costs in the pharmaceutical market.

6.3. Event study

Fig. 5 provides a visual representation of dynamic treatment effects using four staggered DiD estimators: the TWFE model, Borusyak et al. (2024), Callaway and Sant'Anna (2021), and Gardner (2022) estimators. The pre-treatment period estimates, which represent months leading up to the TIS reform, are crucial for testing whether the treatment and control groups followed a common trend before the policy was implemented. In all four panels of the figure, the coefficients for the pre-TIS periods are not significantly different from zero, confirming that there is no detectable difference between the treatment and control groups prior to the reform. This outcome supports the validity of the parallel trend assumption, which is fundamental to DiD identification, ensuring that any post-treatment changes can be attributed to the TIS policy rather than pre-existing differences between the groups.

Post-TIS initiation, all models exhibit a statistically significant increase in drug prices, with the effect becoming more pronounced over time. The TWFE model shows a gradual increase in the coefficient estimates following the reform, with price effects peaking around 10 months after TIS initiation. Similarly, the Borusyak et al. (2024) estimator also reflects an upward trajectory, albeit with larger magnitudes. The Callaway and Sant'Anna (2021) and Gardner (2022) estimators indicate a growing impact of TIS, though with some variation in the timing and magnitude of the price effects.

[34] Under the consistency evaluation regulation, only drugs that passed bio-equivalence testing could remain active on the market after 2018, potentially driving up average drug prices as lower-quality drugs were removed.

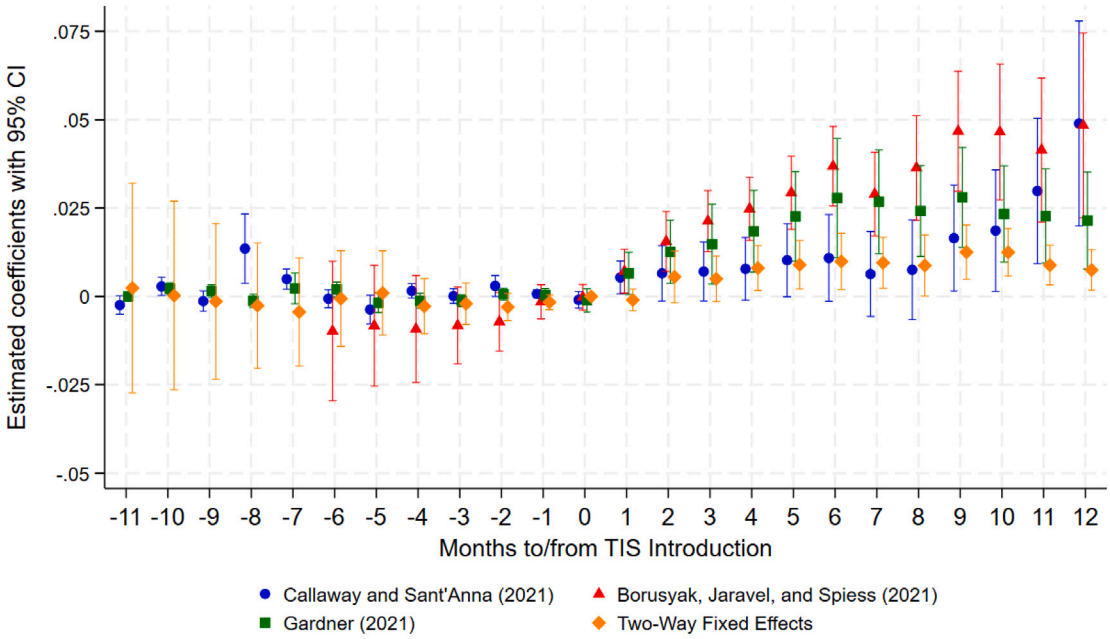

Fig. 5. Event study graphs of TIS's impact on drug prices.
Notes: This figure shows the event study estimated using different methods. The dependent variable is the natural logarithm of CPI-adjusted drug price. Before the TIS reforms, none of the estimates for each period deviated significantly from zero. For the Borusyak et al. (2024) estimates, we do not use all pre-treatment periods, as suggested in Borusyak et al. (2024).

The delayed and growing impact of TIS on drug prices can be explained by the contractual dynamics within the pharmaceutical market. At the time of the TIS reform's introduction, many drugs were still sold at their original prices set in previous procurement cycles. The full effect of the policy only became evident after the expiration of these contracts and the reopening of the bidding process, which led to a reassessment of prices under the new TIS framework. This dynamic effect highlights the importance of considering the gradual evolution of policy impacts, particularly in markets like pharmaceuticals, where long-term contracts are prevalent.

To avoid the confounding impact of COVID-19 on drug prices, we restrict our analysis to data from 2019 and earlier. This implies that our event-study framework captures at most one post-reform year for provinces that implemented the reform in 2018. To extend the analysis, in Appendix B.6, we additionally report two-year treatment effects for provinces that adopted the reform in 2017 or earlier. The two-year event-study results reveal no evidence of persistent price increase over time; if anything, prices exhibit a slight decline in the second year. This finding is consistent with prior literature (von Ungern-Sternberg, 1996; Dobson and Waterson, 1997; Inderst and Valletti, 2011; Hosken et al., 2018; Gaudin, 2018; Rickert et al., 2021), which suggests that greater intermediary concentration does not necessarily raise retail prices. We also estimate Callaway and Sant'Anna (2021)'s event studies by implementation wave (early vs. late) in Appendix B.7, extending the post-treatment window for early provinces where data permit. While each wave shows a post-reform price increase, the early-wave effects are larger and more statistically significant, suggesting greater short-run disruption among early adopters.

In summary, the results in Fig. 5 validate the parallel trend assumption and demonstrate that the TIS reform had a delayed but significant positive impact on drug prices. The increasing effect over time reflects the gradual expiration of pre-existing procurement contracts and the market's adjustment to the new TIS regime.

6.4. Do the results differ across provinces and price levels?

What explains the increase in drug prices following the TIS reforms? While we cannot directly observe changes in distributor behavior due to the difficulty of obtaining financial data from distributors, we explore potential heterogeneity in the impact of the TIS reforms on drug prices across different categorizations: one based on price levels and the other based on GDP rankings.[35] These analyses aim to indirectly shed light on the mechanisms driving the observed price changes. We employ three estimation methods for this analysis: Borusyak et al. (2024), Callaway and Sant'Anna (2021), and Gardner (2022).

As shown in Columns 1 and 2 of Table 5, in the first categorization, drugs are grouped according to their price levels in 2016, with the sample divided into three tiers. We focus on the top and bottom 33% of the sample—representing the highest and lowest

[35] We additionally assess variation in price effects across therapeutic classes; details are provided in Appendix B.9.

Table 5
Heterogeneity analysis: by price level in 2016 and by GDP.

	(1) Lowest 33% price	(2) Highest 33% price	(3) Developed	(4) Underdeveloped
TWFE Estimators	0.009	0.010	0.021**	−0.001
	(0.010)	(0.012)	(0.010)	(0.008)
Borusyak-Jaravel-Spiess (2024)	0.043***	0.015	0.041***	0.029***
	(0.015)	(0.012)	(0.008)	(0.010)
Callaway-Santa'Anna (2021)	0.037***	−0.010	0.017***	0.012*
	(0.009)	(0.006)	(0.005)	(0.006)
Gardner (2021)	0.102***	−0.006	0.092*	0.046
	(0.019)	(0.033)	(0.052)	(0.032)
Observations for TWFE Estimations	41,417	41,736	69,262	79,751
Control Variables	Y	Y	Y	Y
Year-Month FE	Y	Y	Y	Y
Molecule-Province FE	Y	Y	Y	Y

Notes: We triangulate the sample based on the prices of drugs at the time the TIS reform was implemented in the first province. Developed Province: GDP Top 10 Provinces + 4 municipalities; Underdeveloped: Ten provinces with the lowest GDP. All control variables are the same as in Table 3. Standard errors are clustered at the province by ATC major category. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

price levels. The results indicate that for low-priced drugs (the lowest 33% of the sample), the TIS reforms led to a significant price increase. This effect is robust across all models, with the Gardner (2022) model estimating the largest increase. In contrast, for drugs with prices in the highest 33%, the price effects are insignificant and inconsistent. One possible explanation, based on our theoretical model in Section 3 (see Proposition 2), is that for low-priced drugs, distributor markups were already small prior to TIS, and their price elasticities are small, leaving limited room for price reductions as a result of the shorter supply chain. However, the exclusion of less efficient distributors due to TIS may have led to price increases for these drugs.

The second categorization (See Column 3 and 4 in Table 5) examines provinces grouped by GDP rankings, with the top ten provinces and four municipalities classified as "developed" and the bottom ten as "underdeveloped". The results show that, in economically developed provinces—where pharmaceutical manufacturing and distribution networks are more agglomerated—the TIS reforms led to a notable increase in drug prices.[36] This is particularly evident in the Gardner (2022) model, which estimates the largest price increase. In contrast, underdeveloped provinces saw much smaller price increases, with some models showing no significant impact.

These findings are consistent with our theoretical expectations: in less developed regions, supply chains are longer, and markups are higher, and the drug demands there are more sensitive to price changes. For these drugs, shortening the supply chain via TIS is likely to reduce prices. In contrast, drugs sold in more developed regions have relatively small markups, so shortening the supply chain does not significantly reduce prices. In both cases, the net effect appears to be an increase in drug prices, likely due to efficiency losses associated with TIS.

This heterogeneity analysis reveals that the impact of the TIS reforms on drug prices varies significantly depending on both price levels and the economic development of regions. Drugs with lower pre-reform prices and sold in more economically developed regions experienced the largest price increases. These findings highlight the importance of considering price-level and region-specific factors when evaluating the impact of TIS, and suggest that optimizing TIS policy could involve balancing efficiency gains with equitable outcomes across different regions and price categories.

Our analysis leads us to hypothesize that efficiency losses, rather than reductions in markups, may have played a more decisive role in the observed increase in drug prices. This conclusion is consistent with the results from our baseline model (Section 6.1).

While this explanation is in line with our theoretical model, we acknowledge the possibility of other factors not captured in the model that may have influenced the results. For instance, vertical and horizontal integration among pharmaceutical distributors, which may have been encouraged by the TIS reform, could have strengthened their bargaining power. Additionally, a complementary TIS policy that allows for extra invoicing in remote areas—though not fully implemented in provincial guidelines—could have mitigated some of the efficiency losses in supply chains leading to underdeveloped provinces.

6.5. Placebo test

One potential concern regarding our estimates is the possibility that the timing of the TIS implementation across provinces may not be entirely exogenous. Economic and social factors unique to each province could influence the timing of the reform's adoption, which may confound the observed results. Additionally, pharmaceutical companies could have anticipated the staggered rollout of the reform and preemptively adjusted drug prices in response to expected changes. Although our primary regression models control for key economic and demographic variables and address this issue using two distinct time points, we conducted a placebo test to further validate the robustness of our findings.

[36] According to data from listed pharmaceutical companies in the Wind database, in 2016, companies located in the ten provinces with the highest GDP and the four municipalities accounted for 80.3% of total sales and 72.8% of the total number of listed pharmaceutical companies.

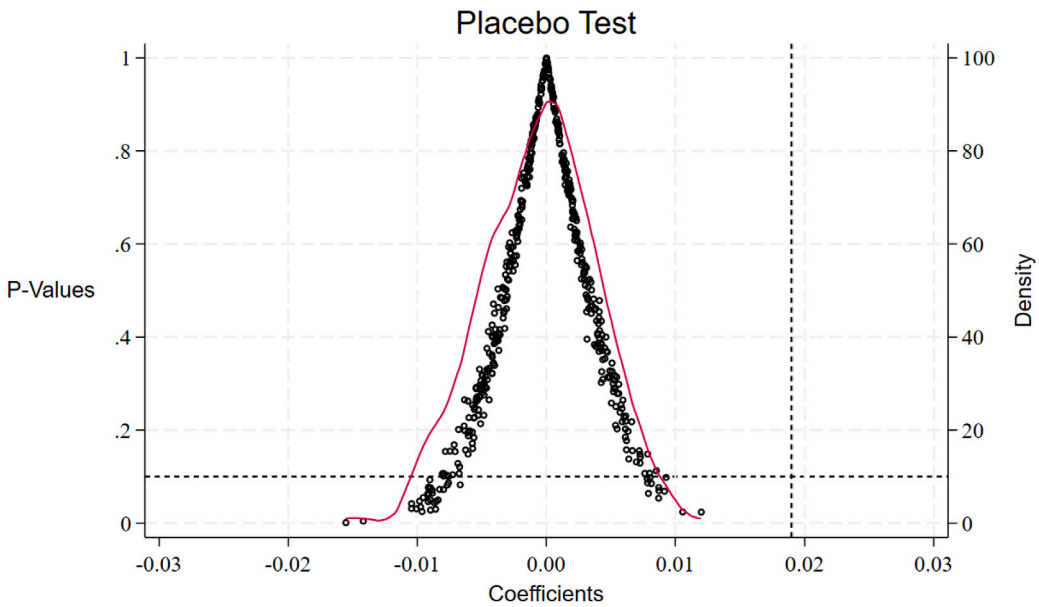

Fig. 6. Placebo test.
Notes: This figure shows the regression results after randomly assigning each province a time to start implementing the TIS and repeating it 1000 times. The dependent variable is the natural logarithm of CPI-adjusted drug price.

Table 6
Impact on procurement distance.

Estimators	Point Est.	Std. Err.	Lower 95% CI	Upper 95% CI
TWFE	0.003	0.008	−0.012	0.017
Borusyak-Jaravel-Spiess (2024)	0.042*	0.022	0.000	0.084
Callaway-Santa'Anna (2021)	0.045***	0.010	0.025	0.065
Gardner (2021)	0.073***	0.021	0.032	0.113
Control Variables	Y	Y	Y	Y
Year-Month FE	Y	Y	Y	Y
Molecule-Province FE	Y	Y	Y	Y

Notes: The number of observations used in the TWFE model is 146,573. All control variables are the same as in Table 3. Standard errors are clustered at the province by ATC major category. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

In this placebo test, we randomly assigned each province an arbitrary start date for the TIS reform and repeated the analysis 1000 times. Fig. 6 displays the distribution of the coefficients obtained from these "virtual TIS" assignments. The resulting distribution is centered around zero, with no significant effects observed in the placebo iterations. This reinforces our confidence that the observed price changes are primarily driven by the actual implementation of the TIS reform, rather than by unobserved economic or anticipatory factors.

6.6. The impact of TIS on procurement distance

The TIS reform limits the number of distributors, potentially excluding more efficient distributors and narrowing the range of pharmaceutical companies available for hospitals to procure from. However, the effect of this narrowing on procurement distance is ambiguous. On one hand, reducing the number of distributors might lead hospitals to procure drugs from geographically closer suppliers, even if their prices are higher. On the other hand, when larger distributors merge with smaller ones, their more profitable partner pharmaceutical companies may be located farther from hospitals, which could increase procurement distance.

To explore this, we calculate the distance between the winning pharmaceutical company and the provincial capital as a proxy for procurement distance, using latitude and longitude data for both. Table 6 shows that procurement distance actually increased slightly after the TIS reform. This rise in procurement distance could contribute to higher distribution costs, further exacerbating the increase in drug prices.

Table 7
Impacts on pharmaceutical firms' financials.

VARIABLES	(1) Sales & Marketing Expenses	(2) Sales & Marketing Expenses/Assets	(3) R&D	(4) R&D/Assets
Post*Treated	331.474***	2.361***	45.535***	3.307**
	(48.696)	(0.864)	(17.062)	(1.664)
Firm FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Province FE	Y	Y	Y	Y
Observations	1315	1315	1,171	1171
Mean of Outcomes	483.101	13.229	103.190	42.110
R-Squared	0.849	0.757	0.807	0.868

Notes: All columns control for firm assets, ROA, leverage ratio, top 5 shareholder ratio, and ROE. Columns 1 and 3 are in millions of RMB, and columns 2 and 4 are in percent. Standard errors are clustered at the firm level. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1.

6.7. The impact of TIS on corporation financials

The TIS reform is expected to have a significant impact on pharmaceutical firms, particularly in terms of their sales and marketing expenses. Prior to the reform, distributors had considerable leverage in the commercial promotion of pharmaceuticals. However, after TIS, pharmaceutical firms are expected to bear more responsibility for marketing and distribution. Our theoretical model incorporates this change, predicting an increase in the sales and marketing expenses of manufacturers (as discussed in Proposition 3).

Table 7 presents the estimated impact of the TIS reform on various financial indicators of pharmaceutical companies.[37] We estimate these effects using Eq. (3), focusing on sales and marketing expenses and R&D expenses. Additionally, we examine ratios such as sales and marketing expenses to assets and R&D expenses to assets to account for variations in firm size. The results reveal significant increases in the level of sales and marketing expenses, as well as the ratio of sales and marketing expenses post-TIS reform. According to our theoretical prediction (*Hypothesis* 2 in Section 3.2), upstream manufacturers are expected to assume a larger share of promotional activities following the implementation of the TIS. Our empirical findings are consistent with this prediction and suggest that manufacturers have indeed become relatively more active—or more effective—in undertaking promotional efforts compared with the pre-TIS period. In this light, it is plausible to conjecture that a meaningful portion of overall marketing activities, potentially including those associated with commercial bribery, may have (partially) shifted upstream along the supply chain rather than being fully curtailed.[38] Although R&D expenses also increased, the interpretation of this result should be made with caution. This is due to other concurrent reforms in the pharmaceutical sector, such as the bio-consistency evaluation, which have also encouraged firms to invest in R&D. Overall, these results suggest that policies targeting the pharmaceutical industry can have spillover effects on R&D investment (Olson and Yin, 2018; Hermosilla, 2024; Chandra et al., 2024).

7. Conclusion

Containing healthcare expenditures, particularly drug prices, is a critical issue in healthcare reform worldwide. This study, through both theoretical modeling and empirical analysis, finds that the TIS reform has led to an increase in drug prices in the short term, with an average rise of approximately 1.9% in provinces where the policy was implemented. This increase is contrary to the policy's intended goals. It is likely attributed to the efficiency losses resulting from the shortening of the supply chains. While there is an opposing effect from the reduction in markups across the supply chain, our empirical results indicate that the overall effect is driven by efficiency losses.

However, we do not interpret the short-term price increase as an indication that the TIS reform has failed. There are two key reasons for this: (1) Supply chain reforms typically require a longer adjustment period. While the administrative reduction in distributors may initially disrupt distribution networks, we expect this impact to diminish over time. (2) The TIS's cap on the number of distributors may curb illicit practices, such as commercial bribery and tax evasion, by reducing multilayer intermediaries and the scope for off-invoice rebates, sham marketing payments, and false-invoice chains. Requirements that hospitals verify manufacturers' purchase invoices and reconcile product-invoice-account records strengthen traceability in drug circulation and may, in turn, reduce the prevalence of falsified (counterfeit) medicines. A feasible quantification of these gains is not possible with the currently available data. Nevertheless, these benefits are real.

[37] The event study figures are shown in Appendix B.8.

[38] Recall that our theoretical result (Corollary 5) suggests that overall promotional activities decline following the implementation of the TIS. In practice, regulatory enforcement against commercial bribery in the pharmaceutical industry has also become more stringent after the TIS, further constraining illegal promotional behavior. However, due to measurement challenges and data limitations, we are unable to empirically identify changes in either the overall scale or the composition (i.e., the share of illegal activities) of promotional efforts. This remains an interesting direction for future research as more comprehensive data become available.

This study has several practical limitations. First, we can estimate only the short-term effects of TIS, using data through 2019 to avoid the confounding effects of China's nationwide roll-out of CVBP since December 2019, which has substantially reduced drug prices to date (Cao et al., 2024; Liu et al., 2024; Jiang et al., 2025). A clean long-horizon evaluation is therefore not feasible given the rapid succession of healthcare reforms in recent years. Second, data on ex-factory drug prices and sales volumes, as well as on the structure of intermediary networks, are unavailable. Access to these data would permit a more comprehensive assessment of the TIS's effects on the distribution sector and facilitate analysis of manufacturers' pricing strategies. Third, as noted in Section 6.4, the TIS reform represents a major reconfiguration of the distribution network. For tractability, our theoretical model does not attempt to fully characterize all mechanisms and tradeoffs—particularly those related to increased bargaining power. Instead, it focuses on the core economic forces most salient in China's context: how shortening the supply chain influences retail prices through reduced markups and potential efficiency losses.

CRediT authorship contribution statement

Xiaoxi Li: Writing – review & editing, Writing – original draft, Validation, Resources, Funding acquisition, Formal analysis, Conceptualization. **Fanyu Liu:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Jianye Yan:** Writing – review & editing, Validation, Formal analysis, Conceptualization. **Nina Yin:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jhealeco.2025.103097.

Data availability

The data underlying this study come from three sources: (i) a leading pharmaceutical industry consulting firm (yaozh.com); (ii) WIND; and (iii) The National Bureau of Statistics of China (NBS, China). The consulting firm and WIND data were used under license and are not publicly available. The statistics from NBS are publicly accessible from the respective government website. Processed analysis files that do not reproduce proprietary yaozh.com/WIND contents are available from the authors upon request, subject to the terms of the data-use licenses.

The authors have prepared a replication package and posted it here: DOI: Li, Xiaoxi; Liu, Fanyu; Yan, Jianye; Yin, Nina (2025), "Has the Shortened Drug Distribution Chain Cut Drug Prices", Mendeley Data, V1, doi: 10.17632/7rrc98cwrw.1.

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